

SQL COMMANDS

DDL • DML • TCL

Creative • Colourful • Easy Revision Notes

DDL — Data Definition Language

CREATE

What it does: Creates a new table.

```
CREATE TABLE Student (ID INT, Name VARCHAR(50), Age INT);
```

Example: A Student table is created.

ALTER

What it does: Modifies the structure of an existing table.

```
ALTER TABLE Student ADD Email VARCHAR(100);
```

Example: An Email column is added.

DROP

What it does: Permanently removes a table and its data.

```
DROP TABLE Student;
```

Example: The Student table is deleted.

TRUNCATE

What it does: Removes all records but keeps the table structure.

```
TRUNCATE TABLE Student;
```

Example: All student records are removed.

RENAME

What it does: Changes the name of a table.

```
RENAME TABLE Student TO Students;
```

Example: Student is renamed to Students.

DML — Data Manipulation Language

INSERT

What it does: Adds new records to a table.

```
INSERT INTO Student (ID, Name, Age) VALUES (1, 'Priya', 18);
```

Example: A new student record is added.

UPDATE

What it does: Changes existing data.

```
UPDATE Student SET Age = 19 WHERE ID = 1;
```

Example: The age of ID 1 is changed to 19.

DELETE

What it does: Removes selected records.

```
DELETE FROM Student WHERE ID = 1;
```

Example: The record with ID 1 is deleted.

TCL — Transaction Control Language

COMMIT

What it does: Permanently saves transaction changes.

```
INSERT INTO Student VALUES (2, 'Arun', 20); COMMIT;
```

Example: The inserted record is permanently saved.

ROLLBACK

What it does: Undoes changes that have not been committed.

```
DELETE FROM Student WHERE ID = 2; ROLLBACK;
```

Example: The uncommitted deletion is cancelled.

SAVEPOINT

What it does: Creates a checkpoint inside a transaction.

```
SAVEPOINT sp1;
```

Example: A savepoint named sp1 is created.

ROLLBACK TO SAVEPOINT

What it does: Returns the transaction to a specific savepoint.

```
ROLLBACK TO sp1;
```

Example: Changes after sp1 are undone.

■ QUICK REVISION

TYPE	PURPOSE	COMMANDS
DDL	Defines structure	CREATE • ALTER • DROP • TRUNCATE • RENAME
DML	Manipulates data	INSERT • UPDATE • DELETE
TCL	Controls transactions	COMMIT • ROLLBACK • SAVEPOINT

Study Tip: DDL = Structure | DML = Data | TCL = Transaction